

Measuring LLM-Powered Conversational Screen Reader Efficiency for Accessing Complex Visualizations

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Introduction

As Artificial Intelligence (AI), such as Large Language Model (LLMs) advances in recent years, Blind and Low Vision (BLV) users have started to use LLMs for varied use cases, including retrieving information, programming, creative writing, and more (Adnin et al. 2024). BLV users also leverage LLMs to overcome accessibility barriers to access inaccessible information, such as high-level structure and details in both digital and physical documents and images. (Tang et al. 2025)

LLMs have shown benefits in assisting BLV users to access information, such as removing the social pressure of asking people (Penuela et al 2024) and increasing task performance. (Kodandaram et al. 2024) An LLM-powered conversational screen reader can increase user agency and allow users to access detailed information from images, which traditional alt text does not provide. (Phutane et al. 2023)

However, BLV users also face significant challenges when using LLM tools. (Adnin et al. 2024; Tang et al. 2025) LLMs can hallucinate by producing outputs that contain wrong information (Adnin et al. 2024; Tang et al. 2025), and it's hard for BLV users to verify such content, which creates a significant barrier to them. (Tang et al. 2025) This forces BLV users to use various ways to verify LLM outputs, such as asking a sighted person, asking multiple questions to LLMs, or using Google Search. (Adnin et al. 2024; Tang et al. 2025; Penuela et al. 2026)

This review synthesizes current challenges and opportunities that LLMs bring to BLV users in accessing information from complex visualizations and images. Section 1 addresses the limitations of traditional alt text for visual descriptions, the BLV user's need for agency and control over image details (1.1), and the image description needs from BLV users depending on context. (1.2) Section 2 addresses the tension between

LLM capabilities for improving accessibility and the harms, such as biases (2.1) and hallucinations (2.2), that pose burdens on BLV users.

To conduct our study, we want to focus on how BLV users use conversational screen readers to access complex data visualization, because data visualization is one of the most complex forms of imagery. When BLV users use traditional screen readers to access a complex data visualization, they encounter cognitive overload and constant hypothesis testing (Zong et al. 2022). An empirical usability evaluation of LLM-powered conversational screen readers in accessing complex data visualizations is needed to evaluate the potential benefits and drawbacks.

1. The Deficit of Traditional Alt-Text and the Demand for Agency

1.1 The need for User Agency and control for accessing information from images and visualizations

BLV users have diverse habits of accessing data from images and visualizations. They want the agency to control what kind of information they want to access from visual data, instead of the author's default (Thompson et al. 2023). When accessing data visualizations, information access behavior varies among individuals. Some users start from a high-level summary insight and explore fewer insights, while others “dug all the way down” because they want their own exploration and give their own conclusion (Thompson et al. 2023). When accessing a static image, BLV users also want different levels of detail depending on individuals and context (Phutane et al. 2023). When BLV users access information from an image, they cannot expect themselves to know what exact information they want from an image (Penuela et al. 2024). For example, one participant in the study by Penuela et al. (2024) expressed that “In one instance [I might want] the background and in another instance it might be a facial expression. Who knows? How would it know what I want?” All these cases reflect BLV users’ varied information access needs and habits, and traditional static alt texts alone cannot satisfy BLV users’ needs for accessing visualizations and images (Jung et al. 2022).

This is because when web content in recent decades becomes “scattered, visual, and dynamic,” screen readers remain the same and do not catch up with their evolution (Phutane et al. 2023). Therefore, a new way is needed for interacting with complex web content. Phutane et al. (2023) find that a hypothetical conversational screen reader allows the user to “dig into” the details of an image beyond an alt text, and satisfies

users' needs in their agency and control over accessing the information from an image. One BLV participant in the study by Phutane et al. (2023) commented that "I would love to have all that information" and "this screen reader sounds like I wouldn't be left out," when hearing the concept of a conversational screen reader.

Recent screen reader products that enable LLM-powered conversational interfaces include TalkBack 16.0 (Al-Heeti, 2025) and NVDA with the [AI-Content Describer plugin](#) (Temmm, 2025). TalkBack, which is the Android default screen reader, started to have a feature to integrate with Google's Gemini nano, so that users can ask what is on an image (Al-Heeti, 2025). Similarly, an AI content describer plugin for NVDA offers a conversational interface powered by LLMs, allowing the user to power the conversational interface using a specific provider's API, or to utilize the built-in, free GPT-4 integration (Temmm, 2025). But how the LLM-assisted interface can help information seeking on images, such as complex images or text descriptions for complex visualizations (Jung et al. 2022) for BLV users, and their actual performance, has not been tested yet (Phutane et al. 2023).

1.2 Information needs from an image or visualization depending on context

The type of information that BLV users want from an image description or visualization depends on different scenarios, such as the location and use cases where the image appears (Kreiss et al. 2022; Stangl et al. 2021). The information goal of the user can determine what kind of description the user wants across different images. For example, participants in the study by Stangl et al. (2021) showed that when the goal is to purchase a gift, they want the image description to focus on the primary object of the image. When visiting a news website about work conditions, participants want the image description to focus on the "Activity of people" and "attributes of the work setting" (Stangl et al. 2021). Participants' information-seeking behavior on a data visualization also depends on the user's specific task (Zong et al. 2022). For example, if the task is related to population growth in the past, they would start from encodings, but if they want to look for sudden changes in population, they would start from the annotation (Zong et al. 2022).

Both Stangl et al. (2021) and Jiang et al. (2024) expressed that "one-size-fits-all" fails to meet the user's needs. While Stangl et al. (2021) primarily focused on static images, Jiang et al. (2024) focused on audio description for videos. Similarly, users want specific details and different kinds of information output depending on "the nature of the video

content” (Jiang et al. 2024). For example, in how-to videos, participants wanted action and equipment details, but for understanding scale or structure, users preferred tactile graphics (Jiang et al. 2024).

While a one-size-fits-all model fails the user, Stangl et al. (2021) find that all image descriptions need the following: “name/identity, perceived gender, clothing style, perceived race/diversity, and facial expressions.” This establishes a baseline for image access.

The current guidelines and metrics for visual descriptions, including audio descriptions for video and image descriptions, are not context dependent (Kreiss et al. 2022; Jiang et al. 2024). The current audio description guidelines for videos “do not consider BLV users' varied preferences across viewing scenarios” (Jiang et al. 2024). Similarly, the current metric “CLIPScore” for image descriptions is not context dependent, but what kind of context-dependent metrics precisely “constitute a useful description is still an underexplored question” (Kreiss et al. 2022).

2. LLM Bias, Misrepresentation, and Hallucination vs. Information Access

2.1: Bias, Stereotypes, and Misrepresentations

Although large language models (LLMs) make a conversational UI for screen readers possible for BLV users (Phutane et al. 2023), they also have downsides. Text output from an LLM can contain stereotypes, biases, and misrepresentations toward people with disabilities and create harm (Adnin et al. 2024). The LLMs would “repeatedly present reductive archetypes for different disabilities” (Mack et al. 2024), such as the stereotypical characterization of blind people being “courageous and resilient” (Adnin et al. 2024). As such, both Autistic Self Advocacy Network (2025), and Bennett et al. (2021) expressed skepticism about the LLM-generated content, due to LLM outputs being discriminative, such as “darker-skinned women were more often algorithmically misclassified than lighter-skinned men” (Bennett et al. 2021), and doctors are usually presented as white people, and fast food workers are presented as black people (ASAN, 2025).

Similarly, in AI-generated images of people with disabilities disproportionately overrepresent the Accessible Technology (AT), such as over-representing wheelchairs,

disabled subjects portrayed as sad, lonely, or helpless (Mack et al. 2024). However, there is no clear solution for a better representation of disability, such as how to present disability respectfully without making stereotypes. (Mack et al. 2024)

On the other hand, BLV users desire information such as face, gender, and skin color from an image (Stangl et al. 2021), but LLMs would sometimes refuse to process such information (Tang et al. 2025), which reflects a challenge of how to balance portraying such traits respectfully while avoiding bias. (Tang et al. 2025). Currently, no AI prototype for generating image descriptions describing appearance characteristics specifically on race, gender, or disability has been tested with actual users (Bennett et al., 2021).

2.2: Hallucination, User Frustration, and Trust

Similarly, when LLM generating outputs such as descriptions for images and visualizations, they would present inaccurate information to users, posing a barrier for BLV users to access information and decreasing trust from users (Penuela et al. 2024; Penuela et al. 2026; Tang et al. 2025; Equal Entry 2023).

When BLV users accessing visual interpretation using a custom built application powered by GPT-4o for accessing visual information, such as “capture photos, receive descriptions, and ask follow-up questions via chat”, the LLM produce 22.2% of responses containing false information (Penuela et al. 2026), and sometimes refusing to provide answers for the user (10.8%), despite the user rated high trustworthy “(mean=3.76 out of 5, max=extremely trustworthy)” (Penuela et al. 2026) and high satisfaction “(mean=4.13 out of 5, max=very satisfying)” (Penuela et al. 2026), For follow up questions for text and graphic interpretations, the LLM (GPT-4o) contained more hallucinations, in which “34.6% of follow-up responses containing one or more hallucinations” (Penuela et al. 2026)

LLM hallucination makes information verification a significant challenge for BLV users (Adnin et al. 2024). When encountering inaccurate information, BLV users use “a variety of strategies to test and verify output” (Tang et al. 2025), such as turning to search engines (Tang et al. 2025; Adnin et al. 2024), visiting related websites (Adnin et al. 2024), or asking for sighted assistance from another person, trying another LLM model (Tang et al. 2025), or asking the LLM multiple questions back and forth to verify if the AI is hallucinating (Penuela et al. 2026; Adnin et al. 2024; Tang et al. 2025).

Additionally, many of the LLMs' web interfaces are not screen reader accessible, which further raises the bar for BLV users in accessing the tool. (Adnin et al. 2024; Leporini et al. 2025) For example, users must navigate excessive keyboard commands to locate response text (Leporini et al. 2025), inconsistent heading structures (Leporini et al. 2025; Adnin et al. 2024), and inaccessible upvote/downvote buttons (Adnin et al. 2024). These non accessible web interfaces for LLM further prevent the user from accessing the tool.

Despite the potential harms from hallucinations and Biases, LLM can improve accessibility by significantly increasing information access speed compared with traditional screen readers (Kodandaram et al. 2024), has a high user satisfaction (Penuela et al. 2026), and a potential LLM-powered conversational screen reader remains a user demand (Phutane et al. 2023). What remains unexplored is "an ethical and productive way forward to combat GenAI-related harms without minimizing the progress in accessibility" (Adnin et al. 2024).

3. Study Design

3.1 Model Improvement vs. Interface Gaps

While LLMs are likely to improve in the future, a recent benchmark in 2025 showing that "top models improved to 0.7%–1.5% hallucination rates" on grounded summarization tasks (Elad 2026), but a perfect LLM generating a dense summary of a complex visualization, such as a table with 393 rows, would still overwhelm a BLV user (Zong et al 2022). How BLV users navigate a complex visualization or images by incorporating LLM, and its efficiency compared with traditional alt text with screen readers, remains unsolved. Because such a gap is rooted in interaction between the user and LLM-assisted technology, but not the maturity of LLMs, an empirical test is required for how the LLM-assisted interface can help information seeking on complex images or complex visualizations for BLV users, and its efficiency compared with traditional screen readers and alt texts.

3.2. Research Question

This study proposes an A/B testing for this research question:

Do BLV users using a conversational screen reader access complex data visualization (Group B) gain more complete information and experience a significant reduction in task completion time and information loss, when accessing complex data visualizations, compared to users utilizing standard screen reader navigation (Group A), without sacrificing significant data accuracy?

3.3 Co-Design

To start the study, we want to recruit researchers who are BLV users to co-lead the study. This is to reflect disability justice principle “Leadership of the most impacted” (Berne et al. 2018) that “Nothing about us, without us”. This is important because it can prevent us from solving imaginary problems, creating disability dongles, and unintentionally creating harmful situations (Zong et al. 2022)

3.4 Recruitment

We plan to recruit 20 BLV users who are proficient with the NVDA screen reader, 10 for each group. This number is chosen because it aligns with the established methodological standard for recent peer-reviewed studies investigating BLV users' information access. (N=10 to N=11 for evaluation of prototype screen reader interfaces (Thompson et al. 2023; Kodandaram et al. 2024), a number further supported by Hwang and Salvendy's (2010) meta-study, which determined that 10 evaluators are enough for discovering 80% of usability problems.

We would make the research transparent by telling participants the intent of the research and the location beforehand, and also proactively asking if participants have any needs, such as in transportation or devices, and avoiding using the word “accommodation” to make the human need a special category. This reflects the disability justice principle of “Recognizing wholeness” (Berne et al. 2018), as we want to treat them as complete humans who have needs, rather than mere test subjects. Then we would randomly assign participants to group A or group B. Group A would use a standard NVDA, then group B would use NVDA with an AI content describer plugin.

3.5 Materials and tools

To ensure a valid comparison, the variable is limited only to whether the screen reader is conversational or not. We would prepare a Windows computer that has NVDA installed, but depending on the group assigned, it may have an [AI Content Describer plugin](#)

([Temme, C. 2025](#)) with GPT-4 integration (Group B) or without the plugin (Group A). We want to avoid building a custom application for this study, to reflect the disability justice principle “Sustainability” (Berne et al. 2018) because we want to use the already existing tools that are already freely available to achieve the same research goal, instead of spending a large amount of cost on developing the computer program.

For both groups, the data they access during the test would be exactly the same: a complex visualization from an academic or government dataset. The visualization should be strictly compliant with the [Chartability](#) Guidelines (Elavsky 2022), based on the POUR-CAF principle. The visualization should include a high-level alt text summary, accompanied by an accessible table using HTML table elements, which participants in the study by Jung et al. (2022) overwhelmingly preferred. Group A would access the accessible data visualization with standard NVDA, and Group B would access the same data visualization with NVDA with the [AI Content Describer plugin](#) ([Temme, C. 2025](#)). that enables a conversational interface.

3.6 Procedure

- Before starting, for both groups, researchers would first tell participants that we are not testing the user but testing the product. Researchers would encourage them to use the think-aloud protocol to say out loud what they are thinking. Additionally, for Group B, researchers would show a tutorial on how to use the AI content describer plugin.
- First, participants would be instructed to find multiple specific data points from the complex visualization (e.g., "Find the nearest peak" or "Find where these two lines intersect," as ideated in Thompson et al., 2023). The specific data points that participants would be instructed to find would be the same for both groups.
- Second, participants would be instructed to freely explore the visualization and report what kind of data they are navigating through. The goal is to measure the interface’s capability to support both high-level comprehension and deep dives without cognitive load.
- After task completion, a post-completion survey will evaluate the participants' experience. This includes open questions, such as whether there is anything comfortable or uncomfortable for them. Both groups would also rate their perceived trustworthiness, satisfaction, agency, and control in the system on a 5-point Likert scale. Group B will additionally be asked to retrospectively rate their general experience using traditional NVDA without the AI Content Describer

plugin on these same metrics. This allows researchers to directly measure Group B's ratings of the conversational screen reader against their own baseline perception of traditional NVDA, as well as against Group A's ratings.

3.7 Data collection and analysis

In order to answer our research question, throughout the task, we would collect both quantitative and qualitative data from the participants. Quantitative data include time on task, accuracy rate, information loss rate, and Likert scale ratings of trustworthiness, user satisfaction, user agency, and control. Qualitative data include think-aloud observations and open-ended survey responses.

Quantitative data:

- Time on task: measured in seconds from the beginning of the task to task completion or abandonment. This addresses whether a LLM powered conversational screen reader can help BLV users find relevant data faster compared to a traditional screen reader.
- Accuracy rate: the the number of correct data points reported by the users divided by the total number of data points reported. This measures the degree to which LLM hallucination compromises the information accuracy of conversational screen reader output.
- Information Loss Rate: The volume of missed data points compared to the baseline. This addresses whether a conversational screen reader can help BLV users access richer information from complex visualizations compared with traditional screen readers.
- Survey Scales: 5-point Likert scale scores measuring perceived trustworthiness, user satisfaction, user agency, and control. These address whether LLM hallucination rates hinder subjective trustworthiness of screen readers, and whether conversational screen readers can provide greater user satisfaction, agency, and control compared with traditional screen readers.

Qualitative data:

- Analyze behavior from the think-aloud protocol to identify frustration, cognitive overload, or hypothesis testing. This tests whether the LLM-powered screen reader can reduce cognitive overload or increase frustration due to its hallucination tendencies.

- Analyze user behaviors during the test and post-completion open-ended survey responses to evaluate and identify other emerging usage patterns.
- Thematic analysis is applied across all qualitative data.

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AI statement

Claude 4.6 Opus and reasoning, and Gemini 3 pro are used to aid in planning the assignment, structuring writings, finding relevant research papers, suggesting fixing and polishing grammar and sentences, and identifying if something is missing. But readings, comprehension, synthesis, and writing are done by the author.

Key prompts:

1. I have done basically all the sources, these 6 themes are my own synthesis, how can I produce a research grade level literature review? How about you give me prompts and I answer with these prompts
2. I need to identify more than 20 sources to write a literature review for this
3. is there any further development from conversational screen readers, such as research on NVDA AI content describer
4. For the research study section should I design a research as well, maybe I also need to refer how to design a research? Nelson norman methodology is one of them?

Related AI Instructions throughout output:

5.3. CTRL+F VERIFICATION STANDARD (MANDATORY VERBATIM QUOTING)

Every factual claim derived from a specific text or search must include the exact, original quote and the original source locator.

Continuous String Rule: Quotes must be continuous strings of text copied precisely from the source. Do not alter punctuation, correct typos, or use ellipses (...) if it breaks the exact string match required for the user to execute a Ctrl+F copy/paste search.

Format:

Claim: [Objective statement]

Verbatim Quote: "[Exact continuous text string]"

Source: [Source Name/Author] (URL or Document Locator, e.g., Page X, Section Y)